Team 6: Aarthi Gunasekaran, Anabelle Ahteck, Emily Kim

templ.

It Snaps On and Off!

Problem Statement:

School-age children and adults have reported breaking their eyeglasses and do not have a replacement pair to use until new eyeglasses are ordered and received.

Problem Justification:

Replacing broken glasses can cost hundreds of dollars, even when only one part is damaged. One of the most common places glasses break is the temple piece. ***templ.*** seeks to solve this by creating a replaceable piece for these situations

Design Brief/Evaluation Criteria:

- Entire prototype must be 3D printed as 1 continuous piece and need no assembly
- Prototype must not exceed 3 cubic inches
- Protoype must be able to be attached to the eyeglasses in 3 seconds
- Comfortable to the user
- Can be dropped from 3' onto carpet/2' onto hardwood without damage
- Must be safe when worn

Mockup



Testing Methods:

- Drop tests on hardwood and carpet from 2 feet and 3 feet, respectively
- Timing how long it takes to attach the prototype onto glasses

Testing Criteria:

- Average attachment time is within 3 seconds in 10 trials
- Glasses do not break after 10 trials of drop tests

Testing Results/Conclusions:

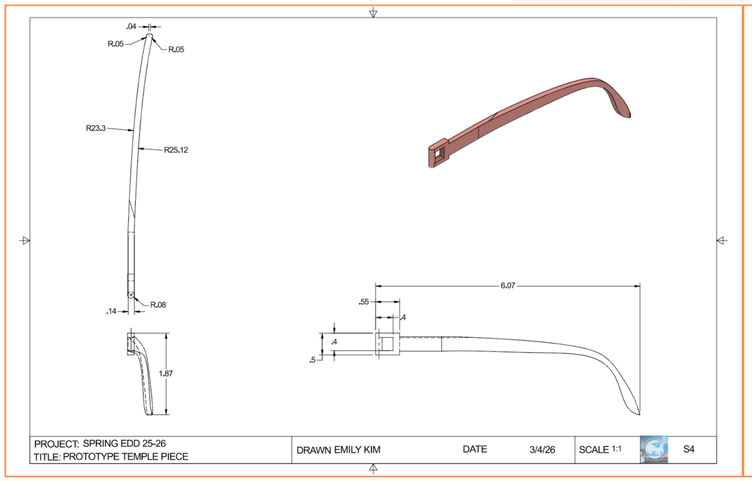
Trial	Time to Attach Glasses (s)	Drop Test Hardwood (Y/N)	Drop Test Carpet (Y/N)
1	1.95	Y	Y
2	1.97	Y	Y
3	1.67	Y	Y
4	1.50	Y	Y
5	2.10	Y	Y
6	1.85	Y	Y
7	2.25	Y	Y
8	1.70	Y	Y
9	1.95	Y	Y
10	2.00	Y	Y

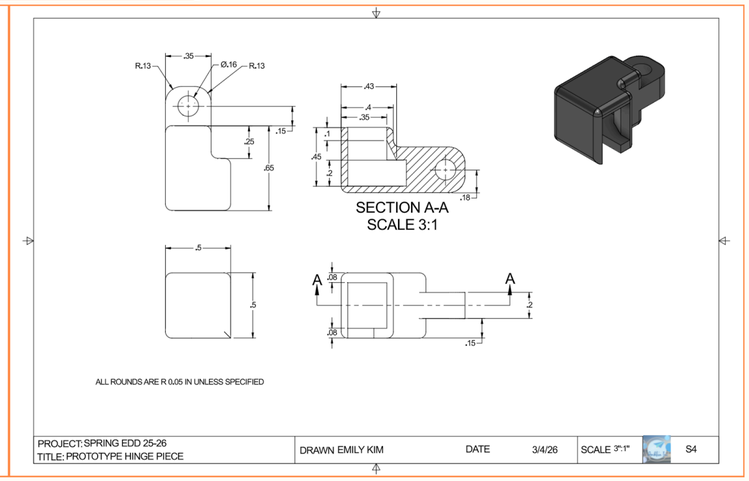
- Range: 1.50-2.25 seconds
- Average: 1.99 seconds

Both range and average attachment times are <3 seconds

The device remained intact and unbroken in all drop tests. It demonstrates a durable design that can withstand typical accidents without breaking.

Technical Drawings:





Bill of Materials:

Item	Unit Price	Quantitiy	z
Polylatic Acid Plastic (PLA)	$0.10 per gram	5.05g	$0.51
Packaging	$0.50	N/A	$0.50

- Retail price: $4.99, which accounts for packaging (plain corrugated mailing boxes can cost around $0.50) and material (PLA costs around $0.10/gram) costs.
- Overall profit: $3.98